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Growing Pains – Lessons on Automation from the Industrial Revolution

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I've found it difficult, recently, to avoid hearing the term "Artificial Intelligence" as it seems the craze over sophisticated prediction models has infected every newsletter in my inbox. Every time I log into LinkedIn, I am bombarded with entrepreneurs developing vague "AI based solutions" that they reckon might change the world. At the same time, it seems like every occupation is a risk of being replaced by artificial humans that can perform any task better, and cheaper, than any lowly meat-bag. Though these fears are understandable, AI threatens the once untouchable domain of human thought upon which the information age was built. Of course, on the other hand, there is the age-old argument that while old jobs become obsolete, new ones we cannot even conceive of will replace them (I just recently learned my roommate works as an "AI prompt writer"). But while we can all look to the future to imagine what careers we might see in the next decade; it might also help to look to the past.

Two hundred years ago, during one of the most critical moments in the economic history of a country that would soon control one quarter of the world's land mass, the quality of life of most of the population plummeted. The ensuing chaos inspired Marx's manifesto and lead the British government to institute the death penalty for those found destroying weaving machines (1). Yet, this period lasted only about fifty years before transitioning to an era of relative prosperity. The question as to what happened may give some insight into what to expect during another period of uncertainty.

At the core of the tumult was rising inequality. The distribution of wealth is an effective way to understand how a market interacts with those within it, while growth will, by in large, improve the livelihoods of every member of a society, too high a concentration of wealth at the top hinders the distribution of resources and can seriously harm those near the bottom. A common way to measure wealth disparities is through incomes. For example, economists point to the increasing difference in incomes for university educated workers compared to those who completed only a high-school degree. This phenomenon is likely the result of automation that can perform the routine jobs usually occupied by the middle class. The remaining careers that cannot be replaced by automation are professionals with very high earnings, or service industry workers who often make no more than minimum wage.

Cumulative growth of real wages by gender and education in the United States (2)

Another mechanism to measure inequality, which we will explore throughout this paper, is by measuring the difference between productivity growth and real wages (wages adjusted for inflation). In a reasonably equitable society, one can expect that (at least) their wages will grow with the economy. If the firm is producing more output at little extra cost, that growth should be reflected in its workers earnings as it can now afford higher wages. Not only that, but if the economy as a whole is expanding, then more goods will be produced, increasing supply and lowering prices. All of which should improve a worker's purchasing power. However, this is not always the case, which suggests something is disrupting the natural path of economic development.

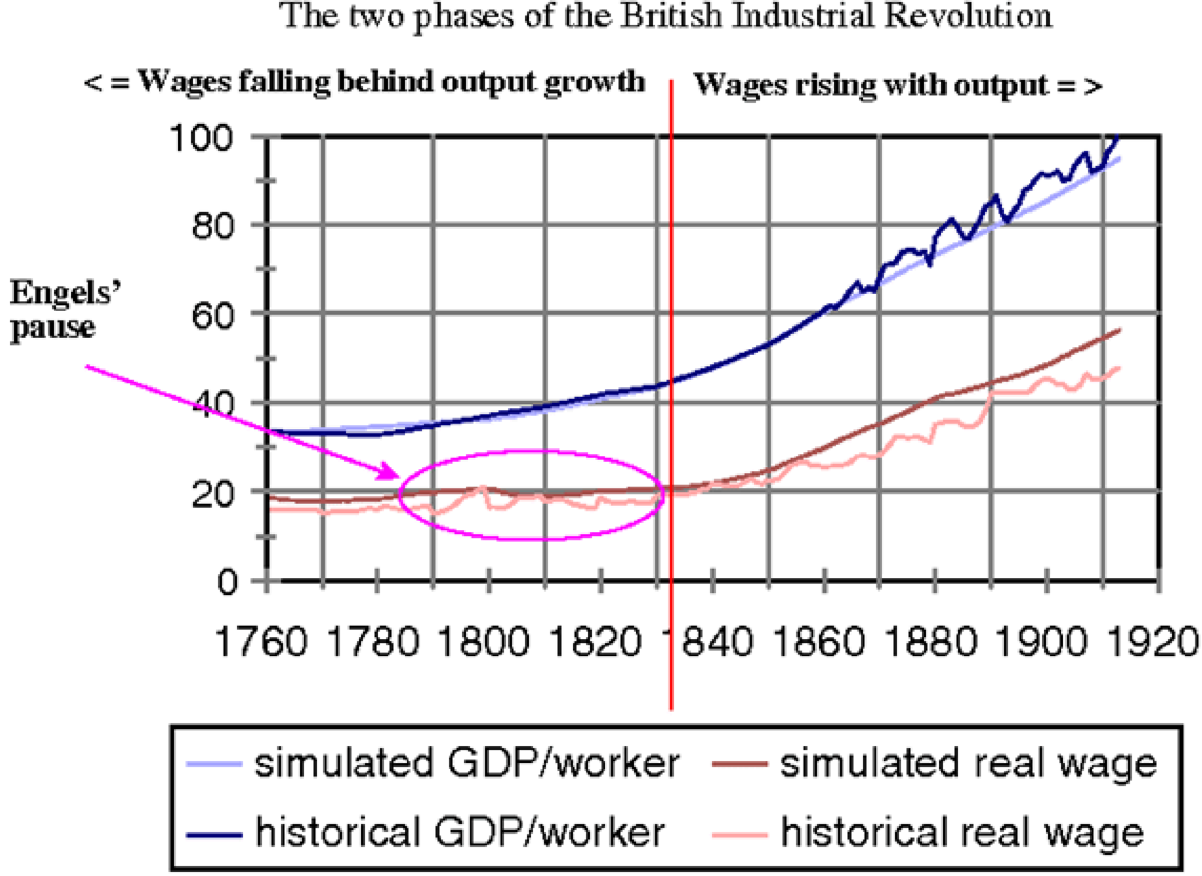
In Canada, between 1976 and 2019 labour productivity grew by 1.10 per cent per year, while the median wage grew by only 0.14 per cent per year (3). While the paper from which this data was pulled cited other factors, namely globalization and de-unionization, others point to a different cause: automation. In a 2021 working paper, Acemoglu and Restrepo found that 50 to 70 per cent of changes to US wage structure between 1980 and 2016 occurred in sectors undergoing significant automation (4).

If productivity increases faster than real wages, it means that the profits are being directed elsewhere, usually to the firms' owners. However, it might also mean that a new method has been developed that produces more outputs with less workers. This is what occurred in Britain in the early years of the industrial revolution.

The Engel's Pause

In 1764 James Hargreaves developed the Spinning Jenny, a hand-powered spindle machine that revolutionized the textile industry. Prior to the Jenny, fabrics were woven by hand in artisans' homes. To increase production, the artisan must increase the number of workers, and even still, output was limited by the hand-loom that had been used for centuries. Hargreaves' machine could spin eight threads at once by simply spinning its wheel, meaning more fabric produced by less workers. The Jenny was also quite large, needing more space to work. An artisan could not simply purchase a Jenny for everyone in their family, they needed the capital to invest in the upfront costs, and new buildings to house their machines. Thus, the factory was born.

The Jenny was one invention credited with kicking off the industrial revolution in Britian. An era in which concentrated centres of production (factories) utilized innovative technologies and processes to output more goods than would have been possible under the cottage system of production. It was during this period that Friedrich Engels, co-author of the *Communist Manifesto*, observed an increasing disparity in the wealth between labourers and the capitalists they worked for in his 1845 treatise *The Condition of the Working Class in England*. The better known of the pair, Karl Marx, suggested that this trend resulted from the proliferation of labour-saving technology, such as the Spinning Jenny, which saved on costs and decreased demand for workers, reducing their bargaining power and leaving them to settle for lower wages. Yet, interestingly, by the time Engel's published his book, real wages began to climb along with output per worker. Thus, with the benefit of retrospect, Robert Allen termed this moment the "Engel's pause" a brief period in which wages stagnated while productivity continued to rise.



Allen's two phases of the British Industrial Revolution (5)

Allen's approach to Engel's pause looks primarily at the economic indicators, dividing the relationship between wages and productivity into two phases (denoted by the red line on the graph above). The first phase resulted from an increasing rate of return on capital compared to labour. By the mid 19th Century, capital boasted a staggering 20 per cent rate of return, with industries such as construction and commerce achieving up to 27 per cent returns. The rapid returns on investment enabled by more efficient methods of production resulted in capital growing from 20 per cent of the national income in the mid 18th Century to 40 per cent a century later. At the same time, labour's share of the national income nearly halved from its peak of 60 per cent in the late 18th Century. Large concentrations of production also required more supporting infrastructure such as buildings and roads. As a result, firms increasingly invested into capital over labour, diminishing wage growth as fewer jobs depressed reduced bargaining power. Mid-way through the 19th Century, the capital accumulation enabled by higher profits could be reinvested back into the firm, making room for wages to rise in line with productivity growth (6).

The Nature of New Technologies

Carl Frey posits a compatible theory, albeit one that focuses more on the technology and social environment itself. In his view, there are, broadly speaking, two types of technologies. Labour replacing and labour enhancing, with the first period being an era of labour replacing. More specifically, *adult* labour-replacing. Not only were inventions like the Spinning Jenny more efficient, but they also required much less technical skill. In fact, they were so easy to use that a child could use it, and they did. Child labour was so prevalent in the industrial revolution that children (anyone younger than the age of 14) made up half of the textile industry by 1830 (7). Child labour was much cheaper than adults as they could often be paid a pittance of what a standard labourer could expect, and they lacked any bargaining power to improve their pay or working conditions.

It would be in the 1830s that the British Factory Acts passed, limiting working hours for children, and mandating two hours of schooling per day for workers under eighteen (8). While this would not phase out child labour overnight, it began the process of making it more costly to employ. At the same time, new innovations such as the steam loom gained prominence. No longer bound by the limits of human strength, these steam-powered machines became much larger and required more skill to operate. This ushered in the second period, one of labour-enhancing technology. The productivity of adult-operated steam-powered machines finally bore fruit for the workers' whose wages had stagnated. Supercharged by the steam locomotive, factories could leverage economies of scale to sell their goods across the Isles. Not only did expanding production create more jobs for labourers, but larger factories needed an entirely new management apparatus. Accountants, maintenance, and administrators all required skill (and literate) workers who could demand more pay from their employers.

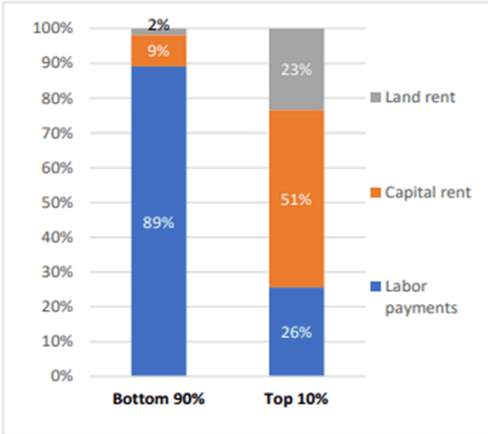
Frey's conclusion from this saga is that the industrial revolution followed a cyclical pattern. Machinery advanced enough to replace skilled labourers so factories could hire whoever would work for the cheapest (which in this case were children). Soon, however, factories could only upscale efficiency through increasingly complex machinery, which meant that skilled workers were once again in demand. Now these workers could leverage this new technology to enhance their skills, with rapidly increasing productivity translating into larger firms that required more and more workers, pulling wage growth back in line with GDP.

Other Factors

There are also other factors which may have caused, or at least exacerbated, the wealth disparity of the Engel's Pause. While they are not especially relevant to the topic of automation, they mirror contemporary events, might paint a fuller picture of how all these forces are acting together today.

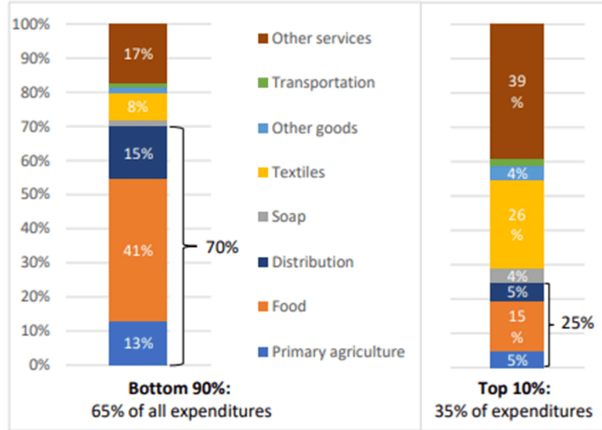
In 1793, outraged over the execution of Louis XVI and concerned over French victories in Germany and Sardinia, the British entered what would be known as the Revolutionary War against the nascent French Republic. By 1801 the French Grand Armée, now commanded by Napoleon, had taken full control of Egypt from the Ottomans. The Napoleonic wars would carry on until 1815, with disastrous effects for the British economy. By the mid 18th Century, Britain had become reliant on grain imports to feed its population, a vulnerability worsened by years of bad harvests by the end of the century. While the relationship between the wars and the price of grain is rather complicated, one of the main culprits in the rising price of goods was the Berlin Decree. After conquering Prussia in 1806, Napoleon declared a state of complete embargo between the continent and the British Isles. Cut off from Europe, the price of wheat rose by 60 per cent between 1807 and 1810 (9).

B. Sources of Income, by class



Source: Based on Lindert (1986, Table 6).

C. Share of Expenditures, by class



Sources: Based on Feinstein (1998, Table 1).

Very little of the bottom 90 per cent's incomes came from rent, while a significant portion of that income was spent on food consumption, so cheaper grain had a significant impact on their quality of life. (10)

Though the war ended in 1815 consumers could not yet catch their breath. That same year, possibly in an attempt to maintain the high grain prices achieved during the Napoleonic Wars, the Tories passed the infamous Corn Laws at the urging of landowning aristocrats. The Laws imposed strict tariffs on the import of cereal grains (including wheat) which pumped up the price of domestic grains. Repealing the Corn Laws in 1846 directly mitigated the growing wealth disparity of the Pause, as the decline in land rents negatively impacted the top 10 per cent of Britons, while reduced prices for goods benefitted the bottom 90 per cent (11).

As the Engel's Pause is based on real wages adjusted that reflect the purchasing power of workers, the political climate that raised the cost of living would have depressed any gains in wages that labourers might have made in a period of rising productivity. This likely sounds familiar to anyone today, as the Russian Federation just recently backed out of a deal to allow grain exports from Ukraine (though the safety net of Canada and Australian grain has so far mitigated the damage to global food prices) (12). While economists debate the exact causes of the diminished quality of life during the Pause, it is likely that, as is the case today, economic and political factors compound to worsen inequality.

Conclusion

Allen's approach suggest that the Engel's Pause is closer to a naturally occurring phenomenon in economics, similar to the expansions and contractions of the business cycle on which macroeconomics is built. The outsized returns on investment granted by new technologies will first allow firms to cut labour costs, reducing wage growth as workers try to hold on to what they can. But the savings made by doing so are then reinvested into production, allowing the firm to grow. That growth then translates into more, better paying jobs.

The available capital may also allow for firms to invest in more complicated machinery which require more skilled workers who are able to bargain for better wages. Though Frey's approach focuses more on the nature of the technology itself, whether it is labour-replacing or labour enhancing. Labour-replacing technology will put workers at a disadvantage as they fight to keep dwindling jobs, whereas labour enhancing technology allows firms to grow, hiring more workers as they do.

In the end, the experience of workers today differs from the experience of workers three hundred years ago. As I mentioned earlier, powerful computers appear to be encroaching on the previously sacred domain of human intellect, making even the most skilled workers obsolete. While the question of whether AI is labour enhancing or labour replacing remains to be seen, we should not resign ourselves to the march of progress. Technology, and the society built around it are human products, and, ultimately, we will always be in control of both.

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